

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 1: Real Numbers

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 HCF and LCM by prime factorisation

1. Two positive integers are $p = 18a^2b^4$ and $q = 20a^3b^2$, where a and b are primes. What is the LCM of p and q ? (a) $2a^2b^2$ (b) $180a^2b^2$ (c) $12a^2b^2$ (d) $180a^3b^4$ **CBSE 2024, 1 mark**

SOLUTION

$$p = 2 \times 3^2 \times a^2 \times b^4 \text{ and } q = 2^2 \times 5 \times a^3 \times b^2$$

LCM takes the greatest power of every prime: $2^2 \times 3^2 \times 5 \times a^3 \times b^4 = 180a^3b^4$

Answer: (d)

2. At a workshop there are 48 French teachers, 80 Hindi teachers and 144 English teachers. Every room must hold the same number of teachers, and all the teachers in a room must teach the same subject. What is the least number of rooms needed? **CBSE 2024, 3 marks**

SOLUTION

For the fewest rooms, each room must seat the greatest possible number, which is the HCF of 48, 80 and 144.

$$48 = 2^4 \times 3, 80 = 2^4 \times 5, 144 = 2^4 \times 3^2, \text{ so HCF} = 2^4 = 16$$

$$\text{Number of rooms} = \frac{48}{16} + \frac{80}{16} + \frac{144}{16} = 3 + 5 + 9$$

Answer: 17 rooms

2 The Fundamental Theorem of Arithmetic

3. Show that $5 \times 11 \times 17 + 3 \times 11$ is a composite number. **CBSE 2024, 2 marks**

SOLUTION

$$5 \times 11 \times 17 + 3 \times 11 = 11(5 \times 17 + 3) = 11 \times 88$$

The number has factors other than 1 and itself (for example 11 and 88).

Hence it is a composite number.

3 Irrational numbers and their proofs

4. Given that $\sqrt{3}$ is irrational, prove that $5 - 2\sqrt{3}$ is irrational. **CBSE 2024, 2 marks**

SOLUTION

Assume, to the contrary, that $5 - 2\sqrt{3}$ is rational. Then $5 - 2\sqrt{3} = \frac{a}{b}$, where a and b are integers and $b \neq 0$.

Rearranging: $\sqrt{3} = \frac{5b - a}{2b}$

Since a and b are integers, the right side is rational, so $\sqrt{3}$ is rational. This contradicts the given fact that $\sqrt{3}$ is irrational.

Hence $5 - 2\sqrt{3}$ is irrational.

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.